

SIH26027: Simple Feature Explanation

1. The Controller Dashboard

What it is: The main control center for the person in charge of the railway section. **In simple terms:** It gives a bird's-eye view of everything happening today. It shows how many maintenance tasks need to be done, how many machines are busy, and a chart showing how much train delay is expected. It highlights "Risks" (like if a task is overdue) so the controller knows exactly what needs urgent attention.

2. Task Management

What it is: A digital to-do list for track maintenance. **In simple terms:** Instead of using paper files, all repair jobs (like fixing a track, repairing overhead wires, or checking signals) are listed here. The system automatically tags emergencies in red, and regular tasks in blue. It also checks if the workers, machines, and materials are actually ready before allowing a job to start.

3. AI Smart Planning (The Brain)

What it is: An automated schedule generator. **In simple terms:** Normally, a human spends hours looking at timetables trying to figure out when to shut down a track for repairs without delaying too many trains. Our AI does this in seconds. You just click "Generate Optimal Plan," and the AI looks at all the pending tasks and available free time between trains. It bundles compatible tasks together (e.g., fixing wires and tracks at the same time) and creates the perfect schedule to minimize train delays.

4. The Train Graph (Visual Map)

What it is: A time-and-distance map of the railway tracks. **In simple terms:** It's a visual chart that plots time across the bottom and stations up the side. The diagonal lines are trains moving between stations. The blue rectangles show exactly when and where the AI has scheduled a track shutdown for maintenance. This lets the controller visually see that the repair blocks safely avoid the trains.

5. What-If Analysis (The Sandbox)

What it is: A simulation tool for emergencies. **In simple terms:** Things go wrong in the real world—machines break, gangs get delayed, or it starts raining. The "What-If" page lets the controller simulate these disasters. If they click "Machine Breaks Down," the AI instantly calculates the ripple effect: it tells them exactly how much money it will cost, how many trains will be delayed, and automatically suggests a backup "Plan B."

6. Analytics

What it is: The performance report card. **In simple terms:** It provides beautiful charts showing how well the railway is operating. It tracks if the repair teams are finishing on time, how efficiently the expensive track machines are being used, and tracks the overall health and completion rate of the different departments.

7. The Mobile App (Field Execution)

What it is: A smartphone app for the workers actually standing on the train tracks. **In simple terms:** Instead of shouting over a walkie-talkie to tell the control room they have started working, the field supervisor just uses this app. It has a very simple 5-step process with giant buttons: **Acknowledge -> Ready -> Start Work -> Complete -> Hand Track Back**. It even works offline if there is no internet on the tracks, saving the data and syncing it later!